

PalmSource®

Conference and Expo

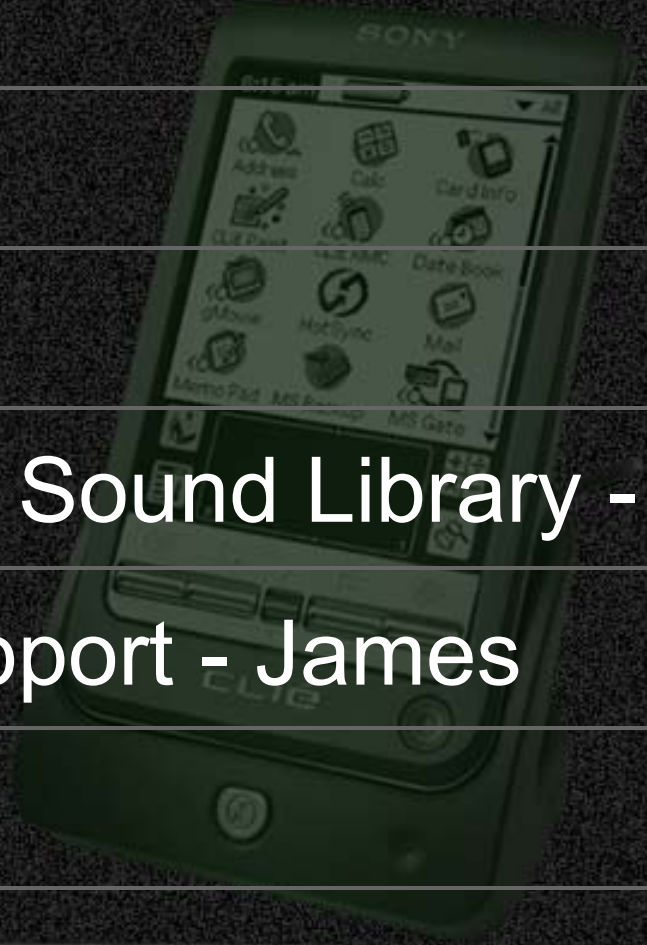


Enabling Your Applications for the CLIE'™ Handheld

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Sony CLIE US Engineering Team

Agenda

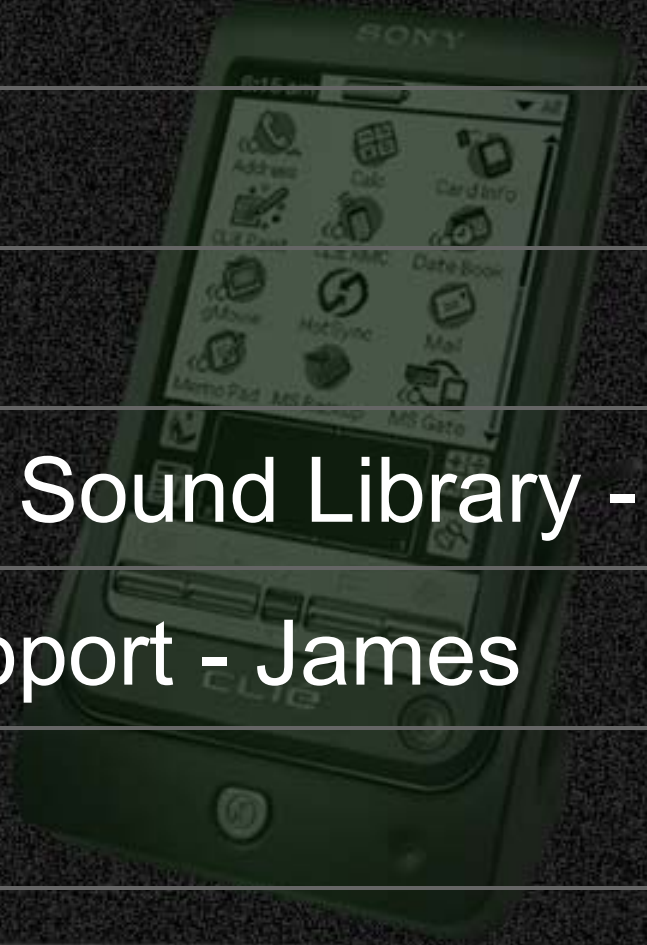


- Introduction – Amy
- Jog Assist - Hsin
- MS Audio and Sony Sound Library - Roy
- High Resolution Support - James
- Q & A

This informative lab session will discuss tools to assist developers to better enable their applications on the Sony CLIÉ™ Handheld.



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- Introduction – Amy
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JogAssist

- JogAssist Functionality
- Preferences Setting
- JogAssist Mask
- Incorporating JogAssist into Applications

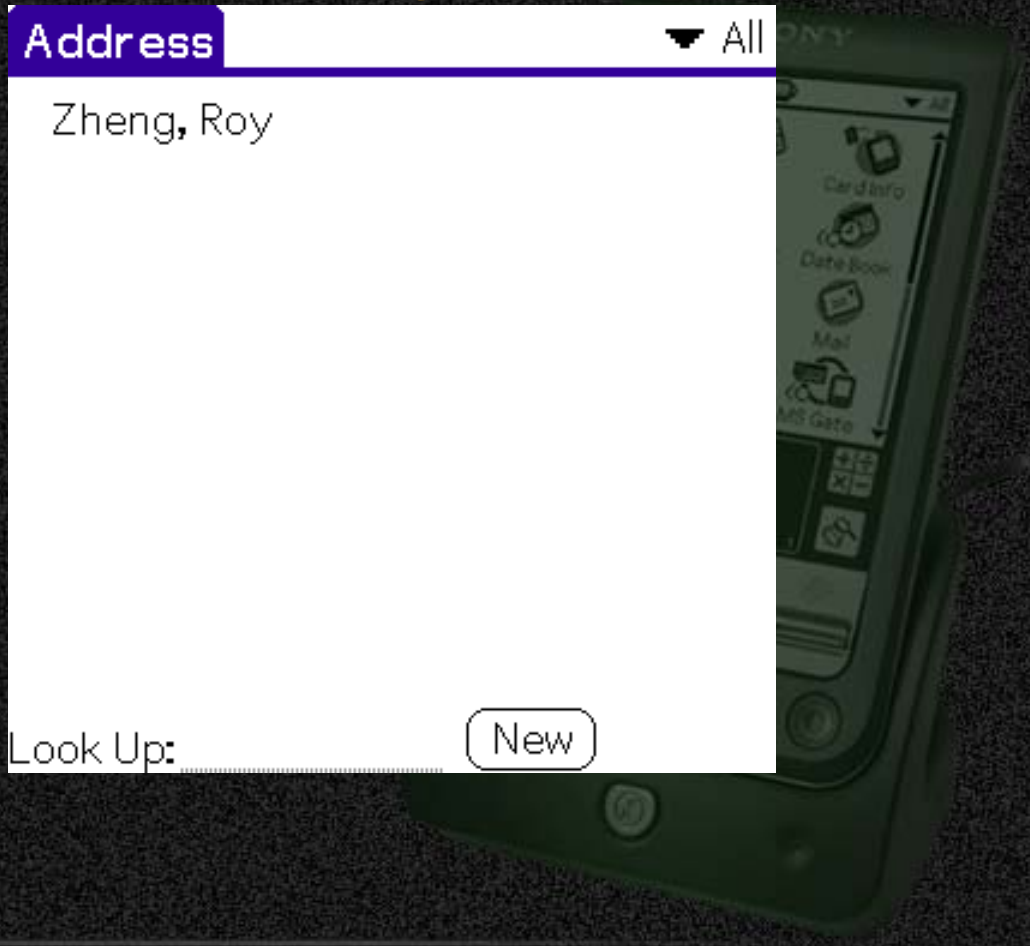


JogAssist Functionality

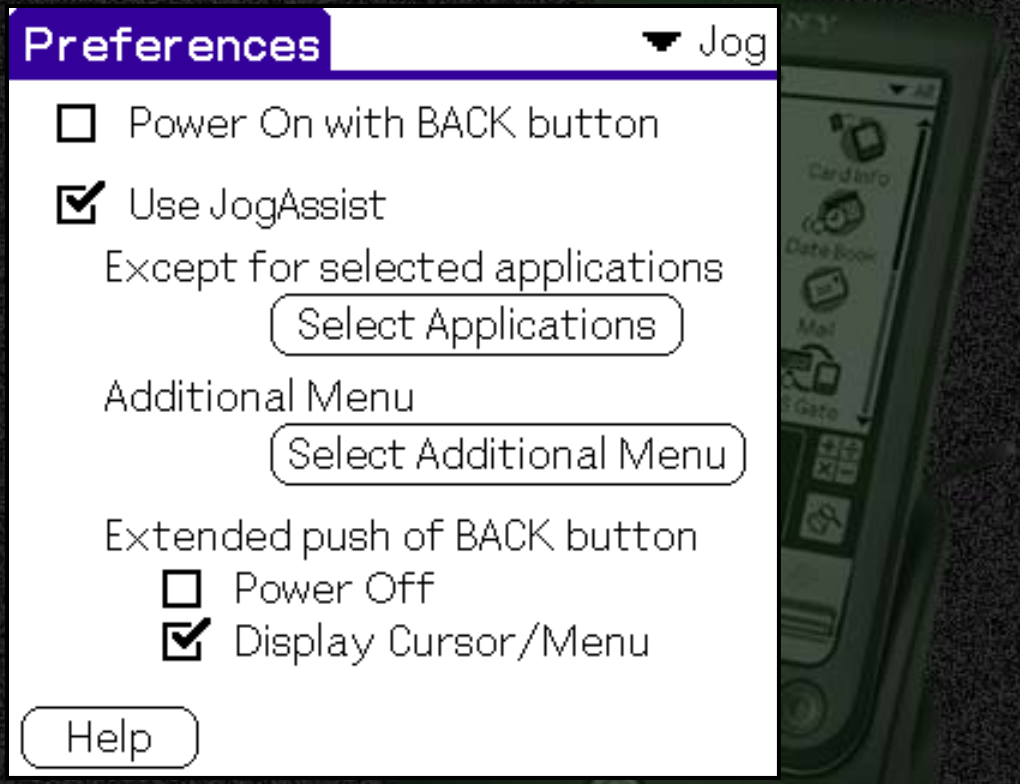
- Enables the use of Jog Dial navigator for applications not supporting Jog Dial functionalities.



Jog Events Processing Example



Preferences Settings

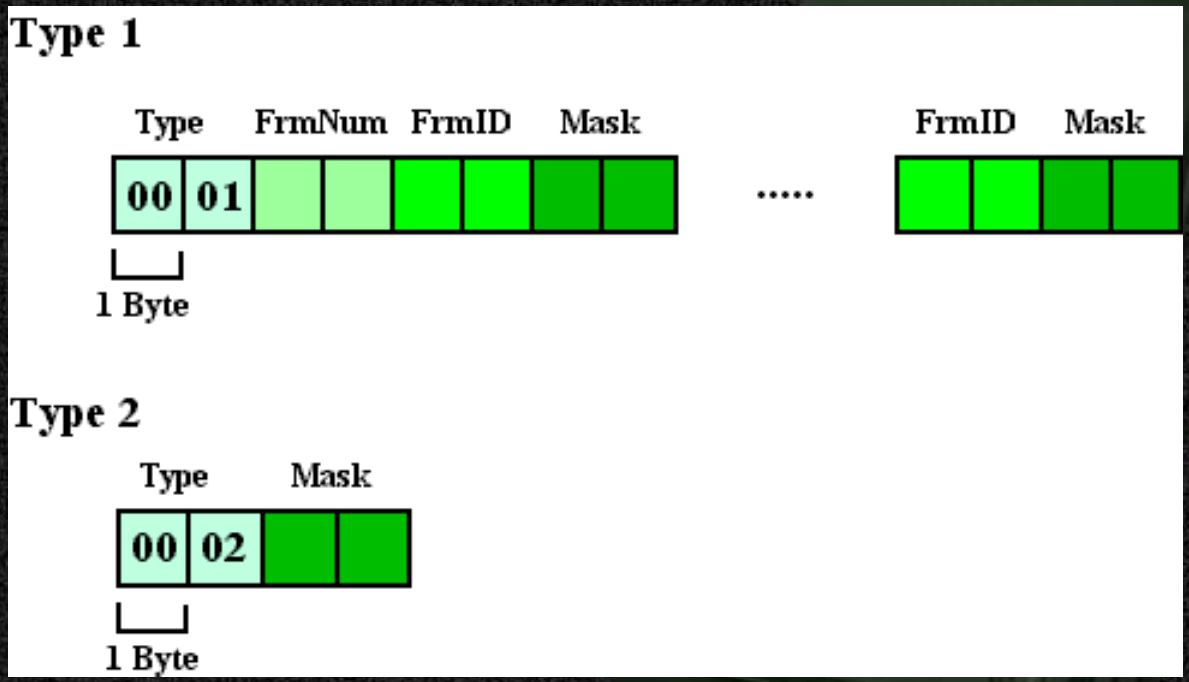


JogAssist Mask

- Restricts JogAssist functionality to prevent JogAssist from interfering with the Jog Dial behavior defined by application.
- See SonyJogAssist.h for definitions.

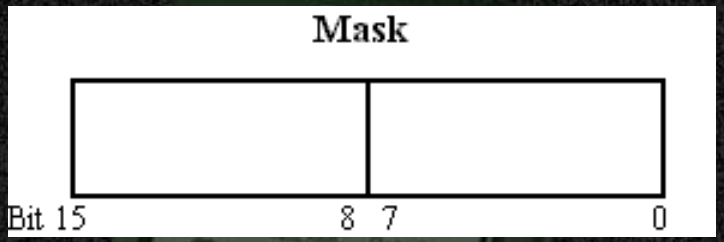


JogAssist Mask Data



Mask Field

Bit 0: vchrJogUp
1: vchrJogDown
2: vchrJogPushedUp
3: vchrJogPushedDown
4: vchrJogPush
5: vchrJogRelease
6: vchrJogPushRepeat
7: vchrJogBack
8 ~ 15: Reserved





Mask Field (cont.)

- 1 means masked (JogAssist function disabled).
- Reserved bits must be set to zero.



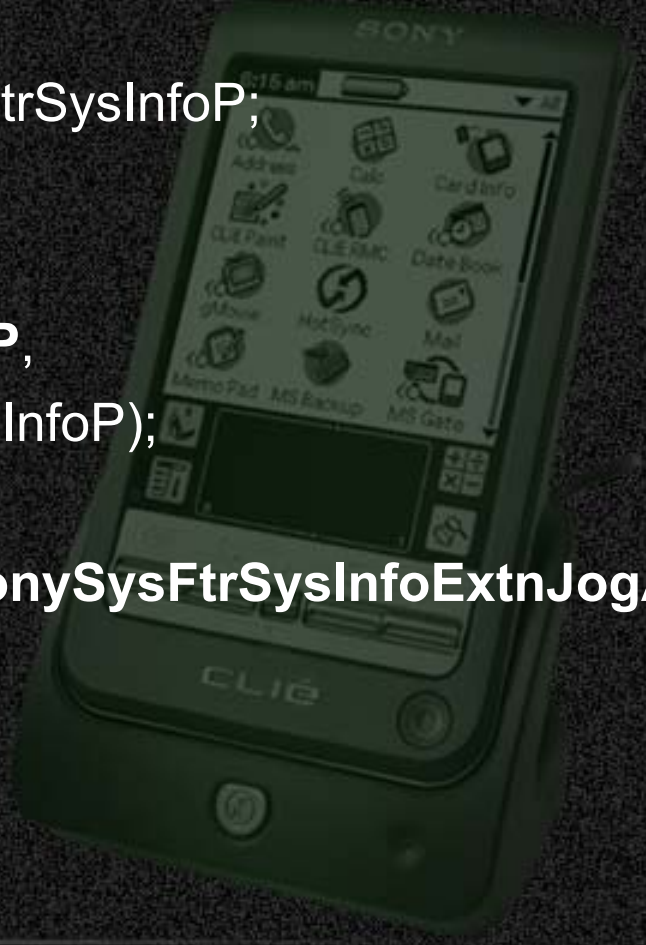
Incorporate JogAssist into Applications



1. Determines if JogAssist is available.
2. Saves old mask and set new mask by JogAssist Mask Pointer.
3. Sets JogAssist Mask Owner (optional).
4. Restores old values.

Determine if JogAssist Is Available

```
SonySysFtrSysInfoP    sonySysFtrSysInfoP;
// get Sony system information
FtrGet (sonySysFtrCreator,
            sonySysFtrNumSysInfoP,
            (UInt32*) &sonySysFtrSysInfoP);
// is JogAssist available?
If (sonySysFtrSysInfoP->extn & sonySysFtrSysInfoExtnJogAst)
{
    .......
}
```

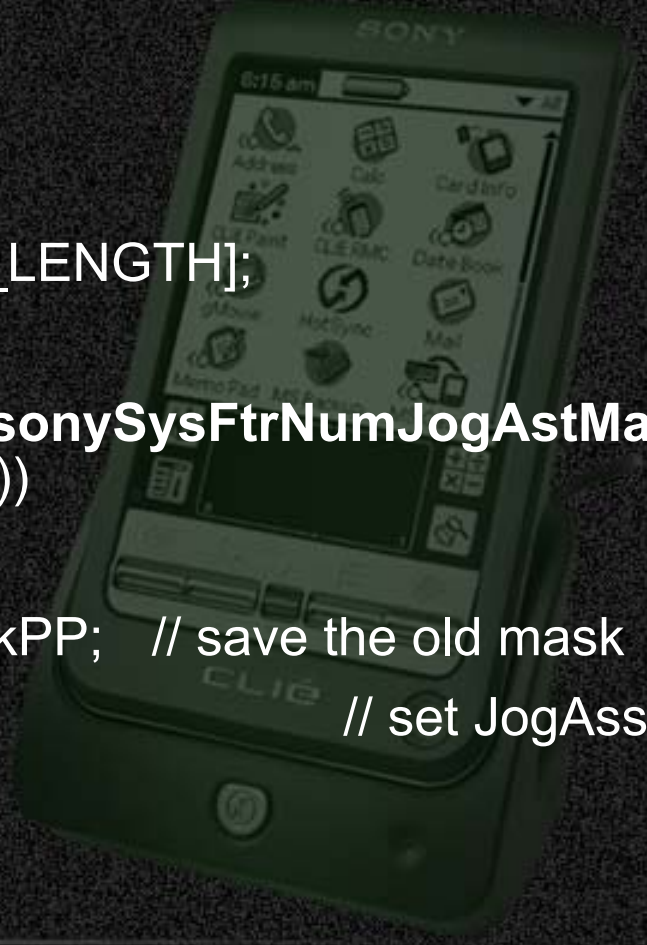


JogAssist Mask Pointer

- Points to the top address of the Mask data.
- Stored in a system-defined address.



JogAssist Mask Pointer (cont.)

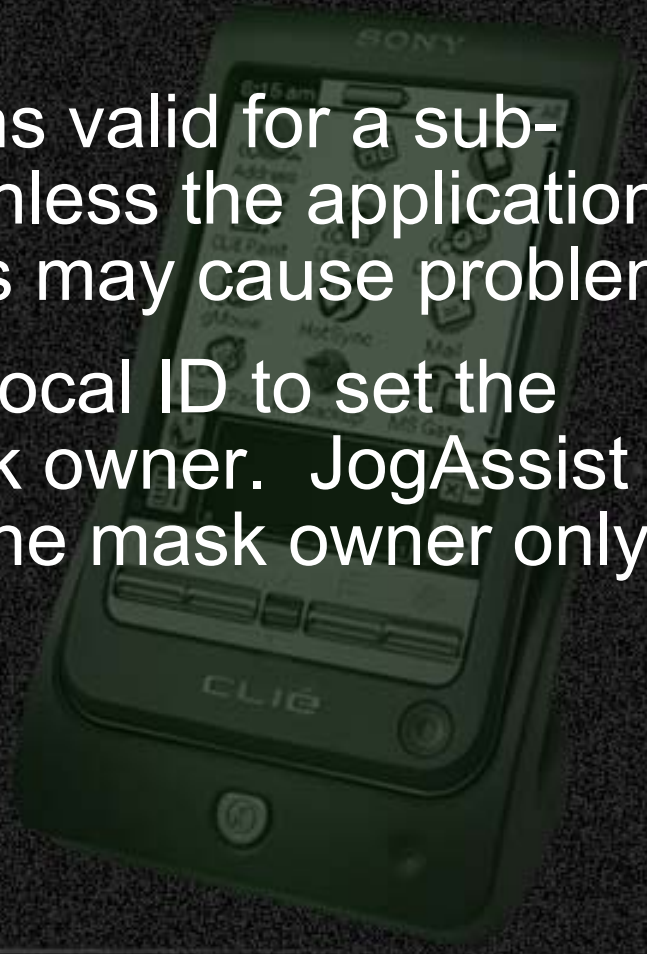



```
UInt16** jogAstMaskPP;
UInt16* oldJogAstMaskP;
UInt16 jogAstMask[MASK_DATA_LENGTH];

 if ( !FtrGet (sonySysFtrCreator, sonySysFtrNumJogAstMaskP,
            (UInt32*) &jogAstMaskPP))
 {
    oldJogAstMaskP = *jogAstMaskPP;   // save the old mask
    *jogAstMaskPP = jogAstMask;                  // set JogAssist
    mask
}
```

JogAssist Mask Owner

- JogAssist mask remains valid for a sub-launched application unless the application sets its own mask. This may cause problems.
- Use card number and local ID to set the application as the mask owner. JogAssist mask is then valid for the mask owner only.



JogAssist Mask Owner (cont.)



```
UInt16 oldFtrCardNo, *ftrCardNoP;
LocalID oldFtrDbID, *ftrDbID;
    ……
 // set JogAssist mask owner
 if (! FtrGet (sonySysFtrCreator,
                    sonySysFtrNumJogAstMOCardNoP,
                  (UInt32*) &ftrCardNoP)
           && !FtrGet(sonySysFtrCreator,
                              sonySysFtrNumJogAstMODbIDP,
                              (UInt32*) &ftrDbIDP) )
{   oldFtrCardNo = *ftrCardNoP;
    oldFtrDbID = *ftrDbIDP;
    SysCurAppDatabase(ftrCardNoP, ftrDbIDP);
}
```

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MS Audio

- Audio Codecs
- Msa Library



Audio Codecs

- ATRAC3
  /HIFI/
- MP3
  /Palm/Programs/MSAudio/



What Msa API Provides

- Basic library operation
- Album and track manipulation
- Audio output control



Basic Operation

- Open/Close
- Background play
- MsaLibEnforceOpen

  sonySysNotifyMsaEnforceOpenEvent



Album and Track

- Album enumeration
- Customize play list
- Playback control
- Track info



Album Enumeration

```
AlbumInfoType albumInfo;

albumInfo.nameP = szAlbumFilePath;
albumInfo.fileNameLength = kMaxFilePathLen;

udwAlbumIterator = albumIteratorStart;
while(udwAlbumIterator != albumIteratorStop) {
   err = MsaAlbumEnumerate(uwMsaLibRefNum,
&udwAlbumIterator, &albumInfo);
   if(err == errNone) {
      //albumInfo.albumtype
      //albumInfo.nameP
   }
}
```



Album Types:

0 - ATRAC3

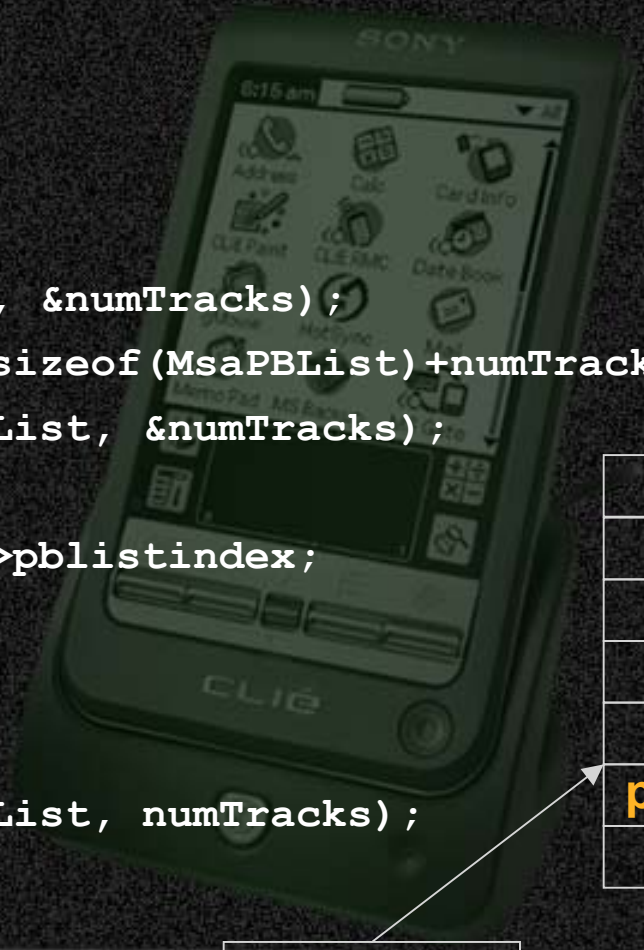
1 - MP3

Customize Play List

```
MsaPBList * gpPBList;
UInt16 * pTracks;

MsaGetPBList(msaLibRefNum, NULL, &numTracks);
gpPBList=(MsaPBList*)MemPtrNew(sizeof(MsaPBList)+numTracks*2);
MsaGetPBList(msaLibRefNum, gpPBList, &numTracks);

pTracks = (UInt16 *) &gpPBList->pblistindex;
pTracks[0] = 2;
pTracks[1] = 1;
numTracks = 2;
MsaSetPBList(msaLibRefNum, gpPBList, numTracks);
```



format
reserve1
creatorID
appInfo
reserve2
pblistindex
…

pTracks

Playback Control

- Direct API call
  MsaPlay/MsaStop
- Use control keys
  MsaSetControlKey



Track Info

```
MemHandle hTrackData;
UInt8 bMask;
UInt16 uwCode;
MsaTrackInfo * pTrack;
UInt16 uwTrackNo;
Char * pchTitle;

bMask = msa_INF_INFALL;
uwCode = msa_LANG_CODE_ASCII;
uwTrackNo = 2;
MsaGetTrackInfo(msaLibRefNum, uwTrackNo, &bMask, uwCode,
   &hTrackData);
pTrack = (MsaTrackInfo *) MemHandleLock(hTrackData);
pchTitle = &pTrack->trackinfo + pTrack->titleoffset;
```



Sony Sound Library

- SMF Playback
  Override system API
- PCM Playback
  Shared library



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Overview:

- Palm's "High-density" API
- Sony's High-res API
- Screen modes
- High-res bitmaps
- Offscreen windows
- Hardware acceleration



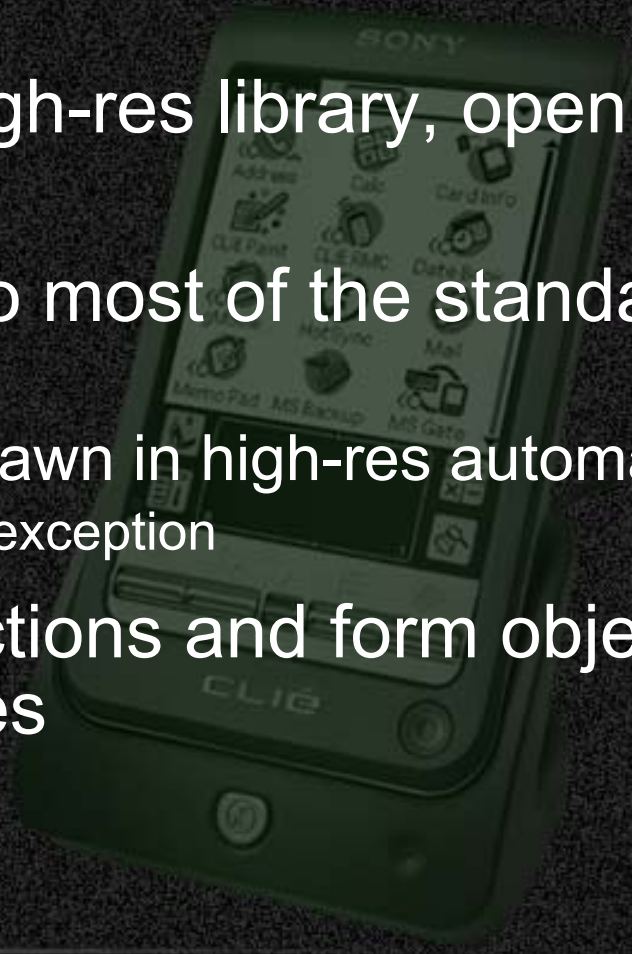
Palm's "High-density" API

- A compatibility layer will allow apps written for Sony's HR API to run on future CLIE handhelds
- Still needs to use Sony's API for existing CLIE devices



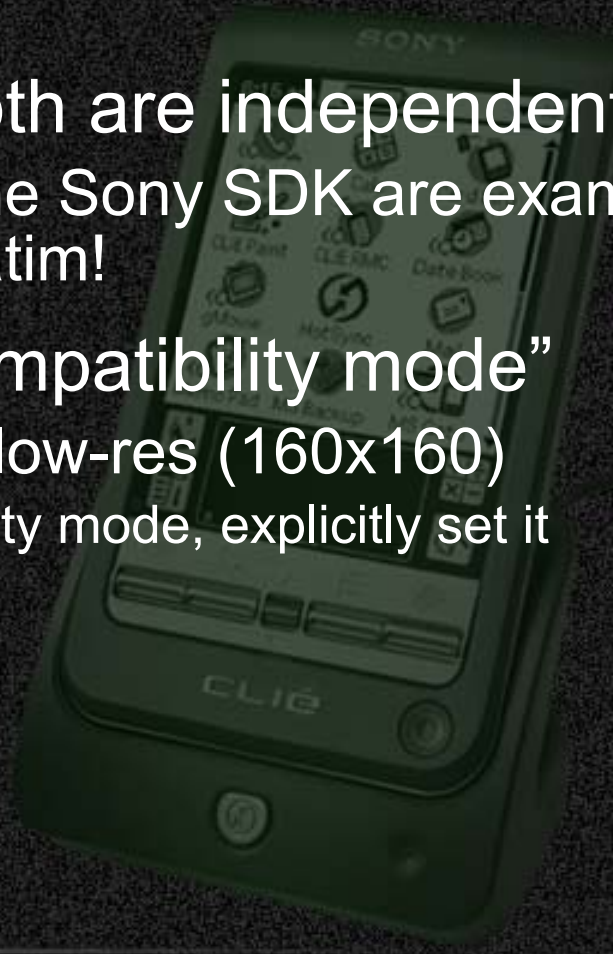
Sony's High-res API

- Check for the Sony High-res library, open it, set the screen mode
- High-res equivalents to most of the standard drawing functions
  - Most resources are drawn in high-res automatically
    - Bitmaps are the main exception
- Standard drawing functions and form objects use low-res coordinates



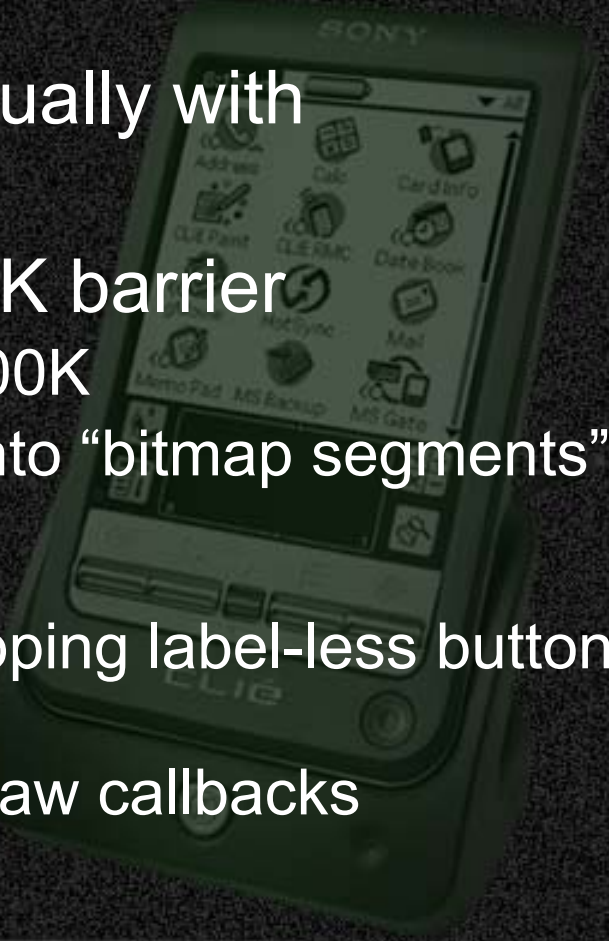
Screen modes

- Resolution and bit-depth are independent!
  - The sample code in the Sony SDK are examples; don't copy them verbatim!
- High-res mode vs. "compatibility mode"
  - Compatibility mode = low-res (160x160)
    - If you need compatibility mode, explicitly set it



High-res bitmaps

- Need to be drawn manually with HRWinDrawBitmap()
- Working around the 64K barrier
  - 320x320 @ 16 bpp = 200K
  - Split up large bitmaps into “bitmap segments”
- Graphical buttons
  - Pre-OS3.5 trick: overlapping label-less buttons and bitmaps
  - Gadgets with custom draw callbacks



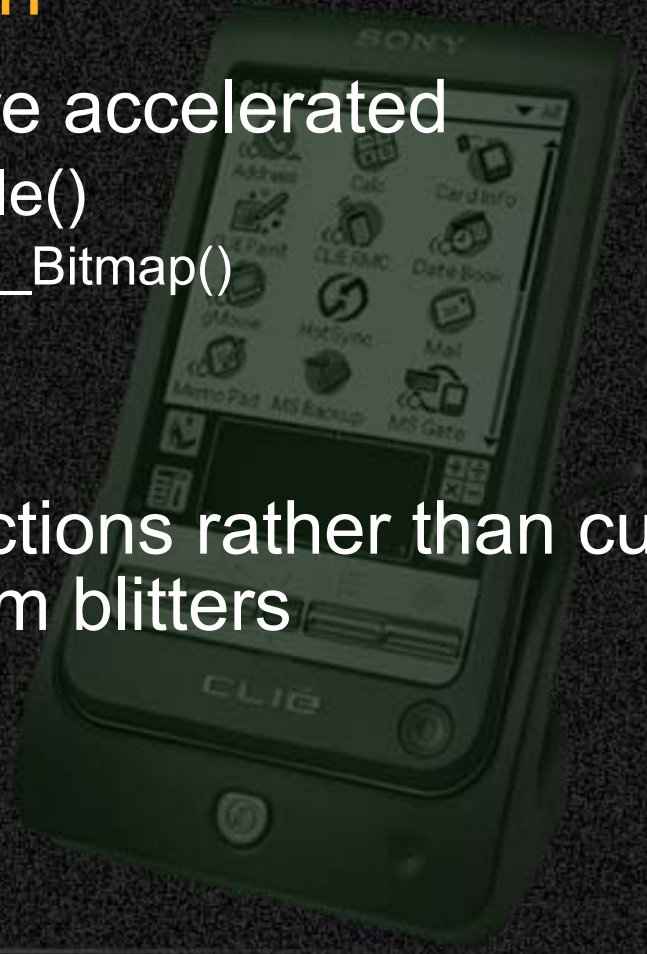
High-res bitmaps (continued)

- Alternative: patch WinDrawBitmap()
  - Allows FormBitmaps and graphical controls to be drawn in high-res automatically
  - Allows drawing “bitmap segments” automatically
  - ***BUT…***



Hardware acceleration

- Some API functions are accelerated
  - [HR]Win____Rectangle()
    - Indirectly: [HR]Win____Bitmap()
  - [HR]Win____Line()
  - WinEraseWindow()
- Try to use the API functions rather than custom drawing code or custom blitters



Q & A





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